Location Tracking for IoT

Capstone Project Report

End-Semester Evaluation

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ABSTRACT

Recently, indoor localization has witnessed an increase in interest, due to the potential wide range of use in different applications, such as Internet of Things (IoT). It is also providing a solution for the absence of Global Positioning System (GPS) signals inside buildings. Different techniques have been used for performing the indoor localization, such as sensors and wireless technologies. In this paper, an indoor localization and object tracking system is proposed based on WiFi transmission technique. It is done by distributing different WiFi sources around the building to read the data of the tracked objects. This is to measure the distance between the WiFi receiver and the object to allocate and track it efficiently. The test results show that the proposed system is working in an efficient way with low cost.

DECLARATION

We hereby declare that the design principles and working prototype model of the project entitled Secure Location Tracking for IoT is an authentic record of our own work carried out in the Computer Science and Engineering Department, TIET, Patiala, under the guidance of Dr. Surjit Singh during 6th and 7th semester (2022).

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Lastly, we would also like to thank our families for their unyielding love and encouragement.

They always wanted the best for us and we admire their determination and sacrifice.

Date: 01st December 2022

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INTRODUCTION

1.1 Project Overview

The Internet of Things (IoT) has become one of the buzzwords in Information Technology. It is an ecosystem of technologies monitoring the status of physical objects, capturing significant data, and communicating that information to software applications through IP networks. The IoT incorporates multiple long-range, short-range, and personal area wireless networks and technologies into the designs of IoT applications. It integrates many technologies, such as Internet, Zigbee, Bluetooth, infrared, WiFi, GPRS, 3G, etc, which can enable different ways to obtain the location information of various objects. Location-based service is a primary service of the IOT. Localisation in indoor positioning systems plays an important role in the IoT.

Location Based IoT applications range from tracking objects and people in real-time, assets management, agriculture, assisted monitoring technologies for healthcare, and smart homes, to name a few. Many localization algorithms and systems have been developed by means of wireless sensor networks for both indoor and outdoor environments.

The interconnection of location information of these devices brings convenience to people's lives, but there is also the problem of leaking location privacy. Therefore, effective management of the location information of the device is crucial to the development of the Internet of Things. In view of such problems existing in the traditional encryption method in the monitoring of device location information, through this project we design and verify the location information storage system of Internet of things devices based on secure storage based on the research of device location of Internet of things and the key technology of secure storage (like block chain).

The easy access and availability of wireless technologies and mobile computing and internet have led to new opportunities in developing mobile applications which purpose is to make people's life easier. Nowadays, a person can possess more than one mobile device intended for different usage such as communication, entertainment, and office work.

The Global Positioning system is the most prominent contribution in determining the position of a user and in routing him to his destination. This system uses satellites to triangulate the location of the GPS device. Though this system has made a good impression in terms of accuracy and is the preferred location-based system for outdoor positioning, when it comes to indoor environments, GPS has proved to be inefficient. The reason for its inefficiency is that in order for GPS to perform a triangulation, the device needs to be in line-of-sight from the satellites. Moreover, the GPS system has a low precision which makes it not suitable for indoor areas. Therefore, when it comes to indoor positioning systems, other alternatives such as Bluetooth, WIFI, RFID and Infrared Red are more preferable.

At present, buildings are equipped with Wireless Local Area Network (WLAN) access points, such as airports, supermarkets, hospitals, museums, etc. It can be effectively implemented to use these access points as a source and to receive WiFi signals to locate user location in these internal environments. However, when it comes to localization of objects indoors, here we need to know more accurate information about the position of such objects inside a building.

There are various indoor localization algorithms based on wireless signals. The main challenges in using these algorithms are to provide acceptable accuracy and deployment cost. For these reasons, a lot of interest has been gained in many research communities using WiFi for indoor localization techniques. Therefore, the proposed method of this paper uses WiFi signals strength to compute the distance between the user's device and the WiFi sender (Access Points (AP) or router). On the

other hand, the problem is that WiFi signals are changing as the obstacles, such as walls, equipment found in the building, and the people around it can affect the signal strengths. The proposed indoor localization technology based on WiFi signal solves this problem by distributing the access point to guarantee the coverage of the whole building. It is simple in implementation and estimation and it can localize the position of a user's device on each floor.

1.2 Need Analysis

With the rapid development in wireless technologies and the Internet, the Internet of Things (IoT) is envisioned to be an integral part of our daily lives. Localization-based services are among the most attractive applications related to the IoT. They are actually, thanks to the deployment of networks of sensors, able to collect and transmit data in order to determine the target's position. A plethora of localization systems are proposed in the literature. These localization systems are based on different positioning approaches, different techniques and different technologies, making them appropriate for some applications and inappropriate for other applications
The position estimation is an essential component of wireless sensor networks applications [1], especially in the environmental monitoring applications such as [2]: animal dwelling place monitoring, water quality monitoring and farming precision. In addition, the location information enables various emerging applications such as inventory management, intrusion detection, road traffic tracking, health monitoring, etc. [2].
The Internet of Things (IoT) is a paradigm shift in the enhancement of the connectivity of the Internet. It provides the technical feasibility for a huge number of sensors, actuators and smart devices to remain connected with it. It adds several emerging services which are essential for connected living initiatives. Among those value added services, localization is one of the prominent one. Localization has a lot of importance in the connected living environment such as security, surveillance and maintenance of law and order. Localization of moving objects leads to tracking which is essential for similar applications. Both localization and tracking are required for emerging applications in several sectors.
Today the students mark the attendance offline on the sheets and there is huge c consumption of papers on a daily basis in the colleges.

1.3 Research Gaps

a. Indoor Localization System Using Wi-Fi Technology:

In this paper, the Wi-Fi signal strength-based object detection and tracking system in the indoor environment was proposed. The proposed system used the WiFi signal strength read from the distributed access points to evaluate the position of the required object. The location is assigned using the MAC address of the smartphone of each user involved. The strength of reading WiFi signals from an access point refers to the location of such a device. The smartphone device is connected to the nearest access point based on the WiFi signal strength, and it is disconnected from the far points. The obtained results showed the accurate performance of the proposed system in terms of allocating objects. It has some limitations like security issues with no security of the stored data and proper management of it.

b. Wi-Fi-based Indoor Positioning System:

The proposed system works in two phases; offline and online phase. The offline phase involves creating a radio map. The radio map stores distributions of RSS values from all detected APs at specific points which are known as marking positions. The marking positions together with the MAC address of each detected APs and their corresponding RSS values will be stored in the database to create the radio map. In the online phase, when the user initializes the mobile application, the sniffer component will collect samples of RSS values from all detected APs. To estimate the position of the user, the data collected will be compared in the radio map. This has limitations like the accuracy of the system can be increased by applying various techniques such as backtracking which will avoid the system from choosing between two close measurements. The system that is proposed and developed in this major project is still in the prototype stage.

c. Location of Things(LoT):

A Review and Taxonomy of Sensors Localization in IoT Infrastructure In this paper, a recent extensive analysis of localization techniques and hierarchical taxonomy and their applications in different contexts is presented. This taxonomy of the localization technique is classified based on the presence of offline training in localization, namely self-determining and training dependent approaches. The future of localization and its implementation for IoT is manifested. Recent localization technologies are analysed and a taxonomical model is presented based on IoT Infrastructure. The emphasis of classification is done based on the presence of offline training state in the algorithms. Various Comparison tables are given for different approaches. How everything can be localized and its futuristic application of IoT-based localization is justified. Different noise distribution and their effect on localization are exposed. Thus the use of localization has different approaches for different scenarios. In this the system has not been implemented and only the efficiency and analysis of various models are done.

d. SECURITY AND PRIVACY PRESERVING OF SENSOR DATA LOCALIZATION BASED ON INTERNET OF THINGS:

The paper addresses the problem of locating the data sensed from the various applications into the cloud, optimizing the energy consumption of storing the data into the cloud and eluding the illegal access by, proper placement of the conflicting and the non-conflicting in the separate host. This process of data locating is based on the non-dominant sorting genetic algorithm-II to arrive at an optimal solution to maximize the quality of service by minimizing the energy consumption, maximizing the resource utilization and enhancing the security by protecting the privacy of the data set by storing the $INDs$ and the NDS in different hosts. The further evaluation using the network simulator-II, provides with the efficiency of the proposed method of data locating compared to the prevailing methods. The paper does not provide the optimization providing the optimal solution reducing the access time of the virtual machines.

1.4 Problem Definition and Scope

The problem of Wi-Fi based indoor localization of a device consists in determining the position of client devices with respect to access points. Many techniques exist to accomplish this, and these may be classified into four main types: received signal strength indication (RSSI), fingerprinting, angle of arrival (AoA) and time of flight (ToF) based techniques. In most cases the first step to determine a device's position is to determine the distance between the target client device and a few access points. With the known distances between the target device and access points, trilateration algorithms may be used to determine the relative position of the target device, using the known position of access points as a reference. Alternatively, the angle of arriving signals at a

target client device can be employed to determine the device's location based on triangulation algorithms. A combination of these techniques may be used to increase the accuracy of the system.

1.5 Assumptions and Constraints

Assumptions:-
1. The user device has Wifi service available and always on.
2. One user can have only one account to access the service.
3. The admin can make unlimited requests and change roles, permissions etc.

Constraints:-
1. The device must reside within the specified region of Localization.
2. The accuracy of the system can be increased by using more Access Points.
3. The development project must be completed within 6 months.

1.6 Standards

● Laptop
○ Model : Inspiron 5490
■ Processor: Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz 2.11 GHz
■ RAM: 8.00 GB (7.79 GB usable)
■ System type: 64-bit operating system, x64-based processor
■ Display adapter:
● NVIDIA GeForce MX250
● Intel(R) UHD Graphics

● Google colab:
Colab notebooks execute code on Google's cloud servers, meaning we can leverage the power of Google hardware, including GPUs and TPUs, regardless of the specifications of the laptop. On cloud server allocated RAM is 12.69 GB RAM and 107.72 GB Disk storage

1.7 Objectives

Localization is utilized in various fields to encourage certain activities e.g security, service provision, navigation and monitoring and tracking. Since the approximate position of objects is the primary thought localization, the objectives talked about give more information about localization.
• Service providing: Patient care in hospitals, security and maintenance of corporate houses, food services in restaurants and hotels are some of the services provided. Security service provision in homes and industries and tracking services to the users are used in static based because sensors are fixed. Technologies such as GPS, infrared, Internet, WiFi, satellite, Bluetooth all contribute to the process of localization to provide services.
• Navigation: It is mainly mobility dependent. Localization in territories, and areas using communication among sensor nodes combined with other techniques has made it possible to navigate and collect data. Exchange of information between various systems likewise information can be exchanged by the use of IoT equipped vehicles and the main system facilitates easy navigation.

• Security: Major and significant difficulties of IoT is security which may be overcome by some secure storage techniques like storing the information on cloud networks. Any device equipped with this system should be able to store the information securely.

1.8 Methodology

In RSSI-based localization systems, the accuracy of the final location largely depends on the range. Therefore, in order to obtain higher accuracy, we need to try various possible methods to improve the range accuracy. In equation (3) we get the relationship of RSSI and distance, where the parameter n means the exponential attenuation factor to distance. In the indoor environment, the external factors deeply affect range value; even in the same location of a node, the value of n may be variable with different circumstances. In order to maintain a certain location accuracy, adjusting the value of n is necessary. Here a method is designed to adjust the range value between blind node and beacon, which depends on RSSI and real distance of beacons. As shown in Figure 5, the real distance between blind node M and beacon node B1 is dm, the calculated distance is m d ′ . The real distance between beacon nodes B1 and B2 is d, while the calculated distance is d′ . RSSI = A - 10nlgd (3) RSSI = A - 10n'lgd' Therefore, we have: logd n = n' x logd The initial value of path loss factor is specified as n′ , and the real distance d can be calculated according to the coordinator of two beacon nodes. d′ is the distance calculated last time.

1.9 Project Outcomes and Deliverables

The proposed solution offers an embodiment to measure and evaluate the fulfillment of objectives. The embodiments to be offered includes:
1. Real time localization system using IoT.
2. Different methods would be analyzed for the said tasks and most ambient one would be incorporated.
3. The user data would be safeguarded using different security protocols and to provide further security it would be stored at decentralized databases using the blockchain technology.
4. We would try to come up with techniques to scale up the demonstrated ideas.
5. Emphasis would also be laid to incorporate a number of other functionalities to make the system diverse.

1.10 Novelty of Work

1. No comprehensive work dedicated to secure Location tracking for IoT using blockchain has been done.
2. Indeed at present general research in this area is still in its infancy,
3. However, work has been carried out to use various algorithms to improve the accuracy but previous works have not comprehensively considered XGboost and Ensemble algorithm for the further improvement and its secure storage using Blockchain

REQUIREMENT ANALYSIS

2.1 Literature Survey

S. No.	Roll Number	Name	Paper Title	Findings	Citation
1	101903311	Manan Ghai	IoT based localization and tracking	In this work, we present the current improvements in localization and tracking using IoT. We start with the differences between the IoT based localization and the non-IoT based systems. Then we show the suitability of IoT for localization and tracking. We show several current applications of IoT in localization and tracking.	Swathi Ramnath; Abhishek Javali; Bhumika Narang; Pallavi Mishra; Sudhir K. Routray IEEE
			Indoor Localization System Using Wi-Fi Technology	In this paper, the Wi-Fi signal strength-based object detection and tracking system in the indoor environment was proposed. The proposed system used the WiFi signal strength read from the distributed access points to evaluate the position of the required object. The location is assigned using the MAC address of the smartphone of each user involved. The strength of reading WiFi signals from an access point refers to the location of such a device. The smartphone device is connected to the nearest access point based on the WiFi signal strength, and it is disconnected from the far points.	Noor Abdul Khaleq Zghair , Muayad Sadik Croock , Ali Abdul Razzaq Taresh IJCCCE
2	101903315	Akshat Singh	Location of Things (LoT): A Review and Taxonomy of Sensors Localization in IoT Infrastructure	In this paper, a recent extensive analysis of localization techniques and hierarchical taxonomy and their applications in different contexts is presented. This taxonomy of the localization technique is classified based on the presence of offline training in localization, namely self-determining and training dependent approaches.	Rathin Chandra Shit; Suraj Sharma; Deepak Puthal; Albert Y. Zomaya IEEE
			Wi-Fi-based Indoor	The proposed system works in two phases; offline and online phase. The offline phase involves creating a radio map. The radio map stores distributions	Arash Habibi Lashkari, Behrang Parhizkar, Mike NG AH Ngan

			Positioning System	of RSS values from all detected APs at specific points which are known as marking positions. The marking positions together with the MAC address of each detected APs and their corresponding RSS values will be stored in the database to create the radio map. In the online phase, when the user initializes the mobile application, the sniffer component will collect samples of RSS values from all detected APs. To estimate the position of the user, the data collected will be compared in the radio map.	ICCNT
3	101903321	Kritarth Gupta	SECURITY AND PRIVACY PRESERVING OF SENSOR DATA LOCALIZATION BASED ON INTERNET OF THINGS	The paper addresses the problem of locating the data sensed from the various applications into the cloud, optimizing the energy consumption of storing the data into the cloud and eluding the illegal access by, proper placement of the conflicting and the non-conflicting in the separate hosts.	Dr. S. R. Mugunthan, ISMAC
			A Novel Approach to Data Storage Using Blockchain Technology	The proposed system currently carries some limitations like the adapter server query data integrity maintenance as the queries itself are not managed on the blockchain. For the initial engine proposed in this paper, naive data structures were used which can end up being computationally expensive in the long run.	Koushik Roy; Nur Islam; Tarango Khan; Mohammad Monirujjaman Khan IEEE

2.2 Software Requirement Specifications

2.2.1 Introduction

2.2.1.1 Purpose

To overcome inefficient work processes, there are various software systems available to speed up the attendance process and reduce manual work. An online attendance management or digital attendance platform is one of them, which is developed to automate daily attendance. Additionally, it helps to maintain accurate records and generate summarized student attendance reports.

The purpose of this management system is to help teachers track student's attendance, absentee record, attendance history & other related documents and keep track of the lectures/labs attended or missed by students. It is the system used to document the time at which students mark their attendance, student's name, roll number, course code and whether they are attending lecture or labs and at which particular venue. Attendance management is done by recording all the above said fields, using our App and thereby recording all the information on spreadsheets on daily or monthly basis, which can be easily accessed by the professors later on for the record.

2.2.1.2 Intended Audience and Reading Suggestions

The intended audience is the engineering class on whom it will be implemented and testing the Attendance Management System. Also, the document is to be utilized by the engineering professor to evaluate the software's design and features.
This document is to be read by the developing team which include 3 members, our mentor of the project and the panel evaluating faculty. The specifications of the project should be read carefully for understanding the overall working flow of the software and its further scopes.

2.2.1.3 Project Scope

This project is able to take attendance of the students and fill their details. The Attendance Management System will allow the teacher incharge to maintain a record of attendance of students in their respective classes from a PC. Also, the system will permit the teacher incharge to maintain all the essential details regarding a particular student. Furthermore, the program provides different modes to edit attendance, view statistics, and take notes on elements pertaining to attendance. The goal is to provide a professor with an easy, portable solution to attendance record maintenance and attendance statistics.
This project can further be enhanced to take attendance of faculty, staff of college to have their details filled and other profile information like salary, personal details etc.

2.2.2 Overall Description

2.2.2.1 Product Perspective

The Attendance Management System is intended to replace the manual model of attendance record keeping by means of roll call and paper records. The roll call and paper records are replaced with a single interaction between the professor and the Attendance Management System. Professors will be able to view details regarding attendance of individual students on their PC and quickly maintain attendance records.
The software is developed for attendance purposes. It is using Api calls for interaction between the frontend and the backend. For frontend and styling purposes we are using Android Studio, JavaScript, ReactJS. In the backend we are using Java along with python and ML models for cleaning the clutter/unwanted data, preprocessing the data, applying feature engineering and presenting the final output in an ordered manner so that it is easily readable by the user.

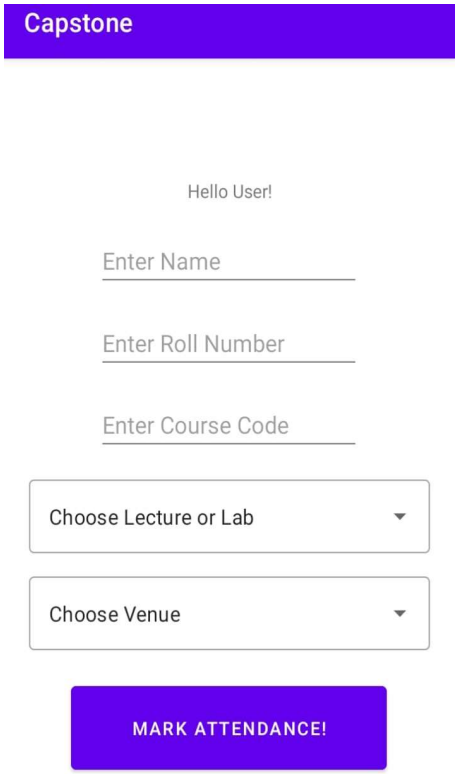
2.2.2.2 Product Features

The main feature of the Management System is that it keeps a record of all essential details of each individual student which can be accessed by the Admin or faculty who can view it later or as and

when required. The app will continuously store the data of the students' attendance whenever an entry is made in the form of a list/dataframe onto the google sheets database. Attendance of each student in every subject would be displayed in this list. Also, the system can further be modified to view pertinent statistics using data analysis on a student's attendance record. Finally, another feature of the Attendance Management System is its flexibility and ease of use.

2.2.3 External Interface Requirements

2.2.3.1 User Interfaces



2.2.3.2 Hardware Interfaces

1. Mobile phones and laptops with active internet connection are required.
2. The systems developed are compatible with 32-bit, 64-bit machines and with the Android Version 9.0 and higher versions.
3. Scanner application to scan the QR code for marking the attendance.

2.2.3.3 Software Interfaces

1. The software is developed for android, ios, windows 7, windows 8, windows 10, etc.
2. The JavaScript library used is ReactJS and AppScripts for the connection between frontend Application interface and the Database.

2.2.4 Other Non-functional Requirements

2.2.4.1 Performance Requirements

The performance of the product made depends on the machine learning model created and the hardware interface developed. Performance benchmarks can be set to judge the output obtained against the input provided. The output desired compared to the output obtained for a given input defines the degree of performance for our Attendance System. The System should provide the desired results for an input:

1. The verification of attendance.
2. The location of the student.

2.2.4.2 Safety Requirements

1. Proper safety measures and precautions must be incorporated for the assembly of hardware components.
2. The hardware components attached should not be opened and tampered with.
3. An adequate power supply should be provided to prevent the system from getting burned.
4. Connecting wires and other hardware materials should be handled with care and connected properly to eliminate any short circuits.

2.2.4.3 Security Requirements

1. Information developed throughout the learning process should be accessible by anyone.
2. Only the admin or the developers of the product should have access to make changes in the model and the dataset. (which consists of student details).

2.3 Cost Analysis

Components	Price	Quantity	Total Price (Rs.)
Wifi Access Point	3000	3	9000

2.4 Risk Analysis

Risk analysis is an important step for any project, to identify any unwanted scenarios for which the level of preparedness is insufficient. While we can predict and reduce the impact of the maximum number of outcomes based on our research and expertise in the subject in hope that it does not have any consequences as the project evolves, it is natural to not be able to cover all aspects of the domain.
As we analyse all the risks and possibilities which can occur while developing the project, it can help us streamline our risks and its impact, improving the chances of a successful project.
The following are some risks that the user should be aware of while using our system:

1. Our System will be unable to function without WiFi.
2. Only the rooms and labs that our system has been programmed to understand will be interpreted.
3. If the requests are longer, the response time is longer, and the system will produce the delayed results.

METHODOLOGY ADOPTED

3.1 Investigative Techniques

The technique we are using to accomplish our project's objective is experimental. With the help of ML Models, and Networking attributes we will classify the attendance of students captured by the app that will enable us to identify the correct attendance marked by the student.

3.2 Proposed Solution

Our proposed solution revolves around the idea of a comparative differentiation between the widely used Wifi localization methods based upon different parameters and its ability to be integrated with a blockchain enabled system for secure storage of the obtained location coordinates. In the last few years research is going on for finding the efficient algorithm for localization. The localization of the sensor node is a two step process of first physical measurement with some reference and second estimate the location from the physical measurement data. Sometimes nodes with limited hardware do not conduct physical measurement rather they try to calculate the position from neighbor node information. The following approaches for localization are existing in the literature. These are based on range measurement, available from anchor nodes, fine-grained, coarse-grained, incremental and concurrent localization methods. The basis of classification of the above scheme is based on either anchor node or range measurement. From the existing literature and IoT application it is found that there is a need for review of existing technology to address the localization problem to the IoT infrastructure. The selection of proper localization methods in the IoT scenario aims to address the restriction and avoid the pitfalls. The proposed taxonomy classifies the localization approach based on offline training. Since IoT infrastructure analyzes huge data and devices. There is a huge scope of application offline training for localization. So the training step is the key for the taxonomic hierarchy of localization methods in IoT infrastructure. The classification is presented in Fig. The localization approach is reviewed into two categories considering the IoT scenario as Self Determining Method Training Dependent Method. We would be performing a thorough analysis of the different localisation algorithms and would finally recommend the best possible solution. From the product point of view we would be building a prototype to demonstrate our findings based on the results of the research. We would be designing an attendance based system that would require the location access of the respective users and mark their attendance based on the access points of a network they are connected to. Our proposed model matches the BSSID of the access point to which the user is connected with the list of permitted BSSID of the access points present in that classroom. Further we would be creating a web interface where a teacher can access the attendance of the students based on their attributes like room no. , course code, batch, day, time etc. and would get the final excel files sent to their email address.

3.3 Work Breakdown Structure

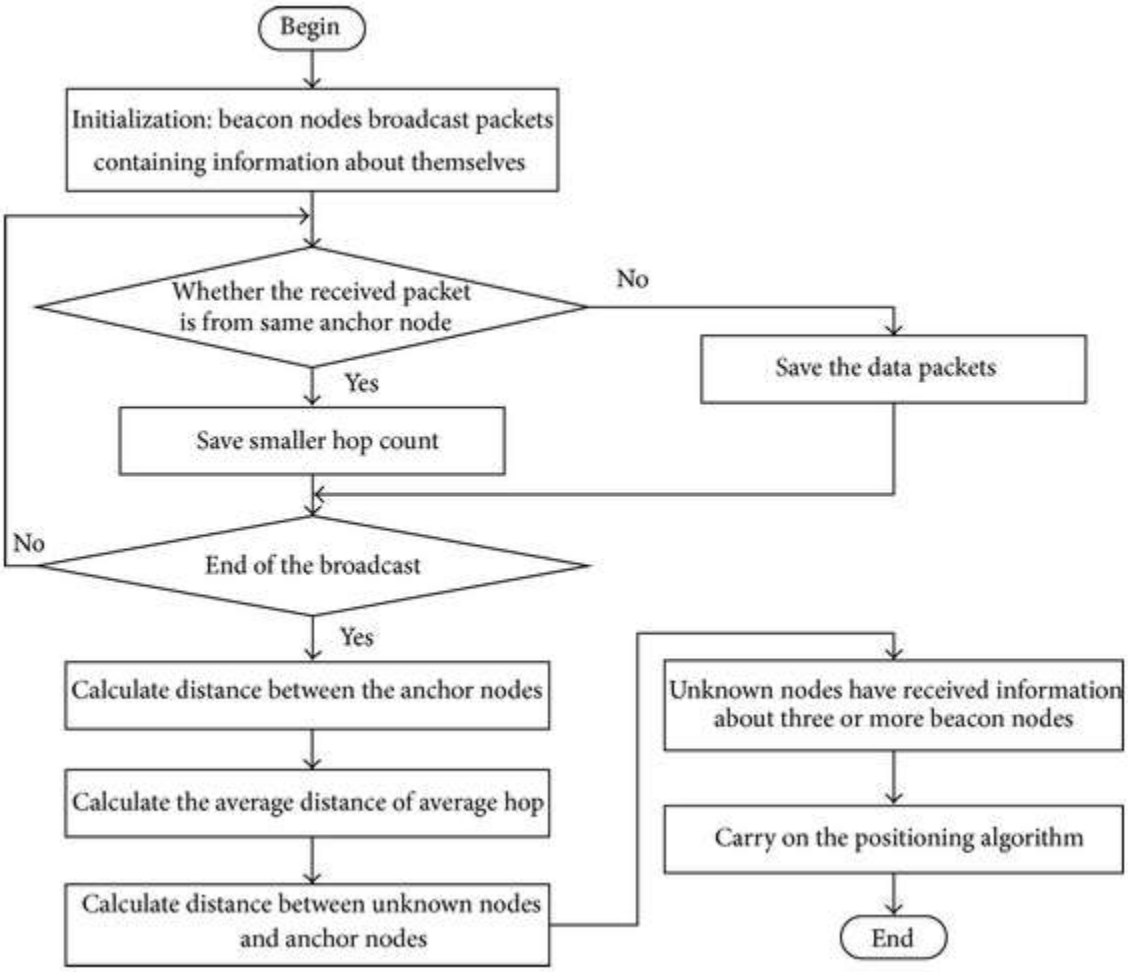
3.4 Tools and Technologies Used

We would be making use of a vast variety of tools and different technologies in our project work which include:

1. Machine learning models.
2. Jupyter.
3. Python.
4. Networking Principles.
6. IOT devices and their functionalities.
7. Android Studio.
8. Java
9. JavaScript
10. ReactJS

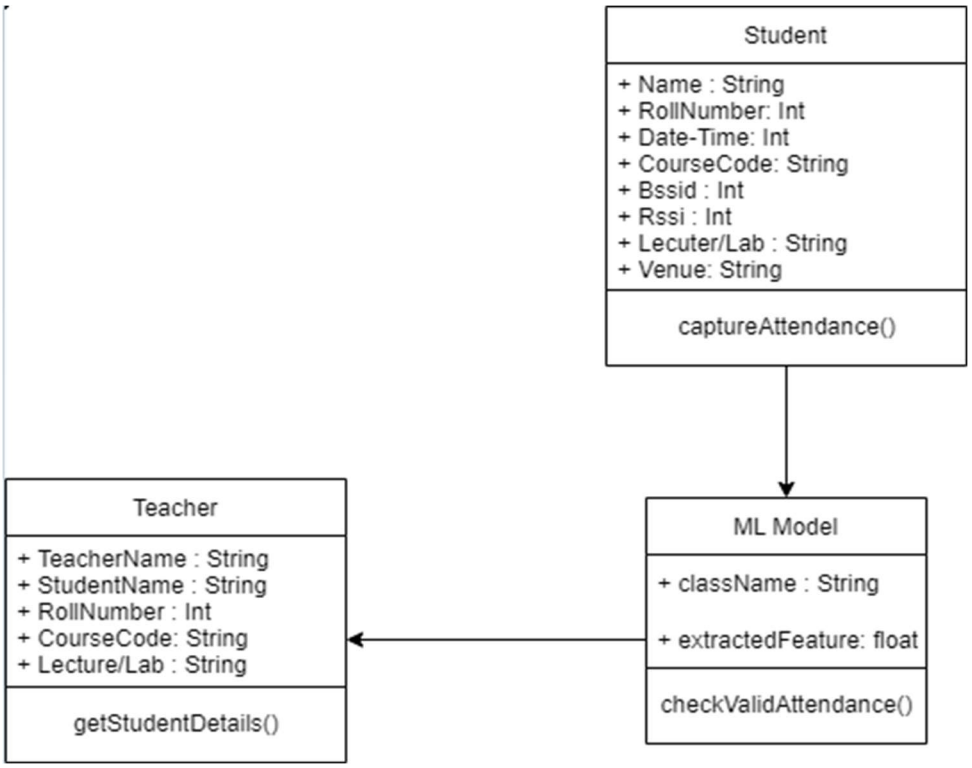
DESIGN SPECIFICATIONS

4.1 System Architecture



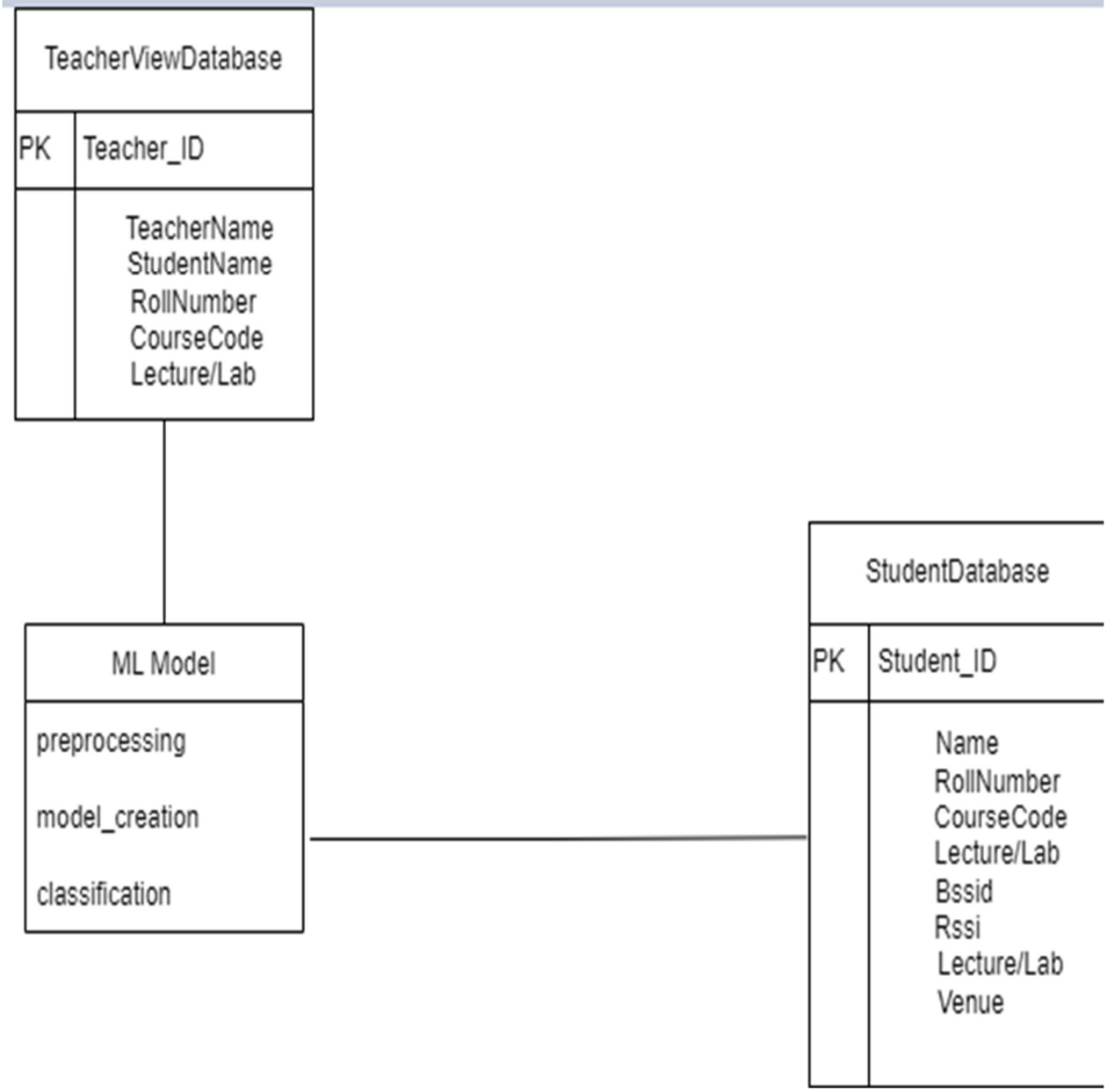
4.2.1 Class Diagram

Our class diagram has three major classes. In the first class, i.e Student class like getting student name, roll Number, Course Code etc. details from the students.



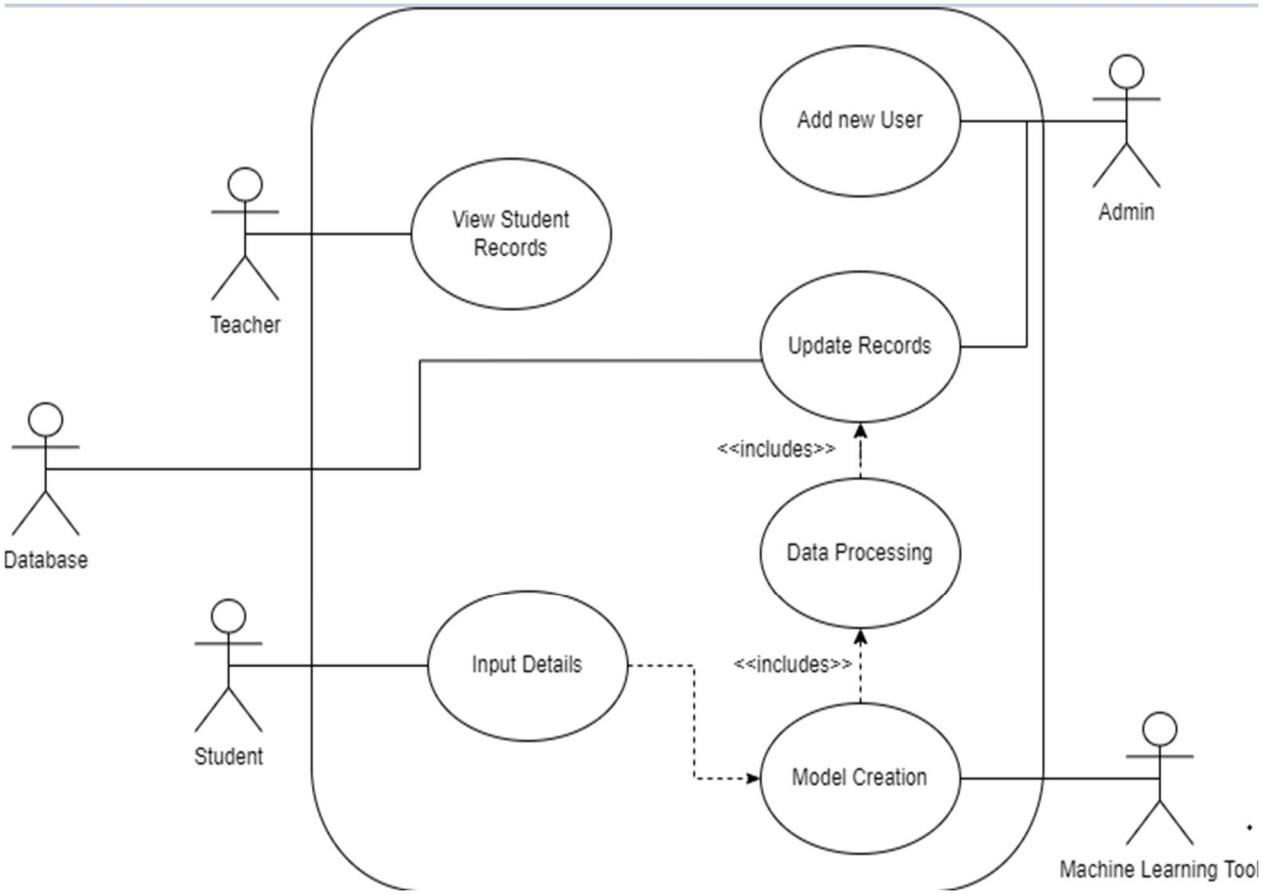
4.2.2 ER Diagram

An ER Diagram is a form of flowchart that shows the relationships between "entities" like people, things, or concepts within a system. They are most frequently used in the disciplines of software engineering, business information systems, education, and research to build or troubleshoot relational databases.



4.3 User Interface Diagrams

A graphical representation of a user's potential interactions with a system is called a use case diagram. It displays different user types and use cases for the system.


Add new User
Admin
View Student Records
Teacher
Update Records
Database
<<includes>>
Data Processing
Input Details
<<includes>>
Student
Model Creation
Machine Learning Tool

IMPLEMENTATION AND EXPERIMENTAL RESULTS

5.1 Experimental Setup

The testing of each feature of the application is an ongoing process. We built the appropriate front and back end and then integrated both of them. We thoroughly tested the applications for a number of functionalities that it offers. We already have a database mapping consisting of the Room Number to the BSSID mapping. We tested the system for both valid as well as invalid or false inputs. There is a flow of information within the application to fulfil the requirements. mentioned as under.

S.No.	Operation Input	Keywords
1.	Login	Username / ID , Password
2.	Marking the attendance	Venue, Class Type, Course Code
3.	Viewing User Information	----------
4.	Downloading attendance sheets	----------

Table No. 5.1 Operations and keywords used in the application

5.2 Experimental Analysis

5.2.1 Data

S.No.	Table	Attributes
1.	User Information	Name , Roll Number , E-Mail , Password
2.	Mapping Table	Room Number , BSSID
3.	Attendance Table	Time , Venue , Class Type , Course Code , Attendance status

Table No. 5.2 Table in the Database along with the attributes

5.2.2 Performance Parameters

For the successful completion of the application all the objectives / features which were the major building blocks of the concept should be working as desired.

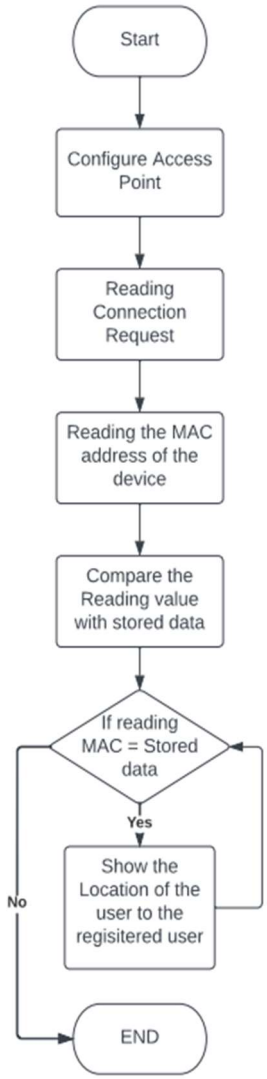
S.No.	Objective
1.	Secure Login for the Use’s
2.	Convenient and hassle free attendance marking
3.	Display all the necessary details on the dashboard

Table No. 5.3 List of the objectives of the application

5.3 Working of the Project

5.3.1 Procedural Workflow

In the below diagram we showed the complete workflow of our project. The first step is to configure the access point. and then the application sends the details with the BSSID captured of the Wi-Fi access point to which the device is connected. Then it compares the recorded BSSID to the configured BSSID of the access point if the value matches then it stores on to the database else it shows that wrong details is provided so that only verified and accurate location of the student is captured.



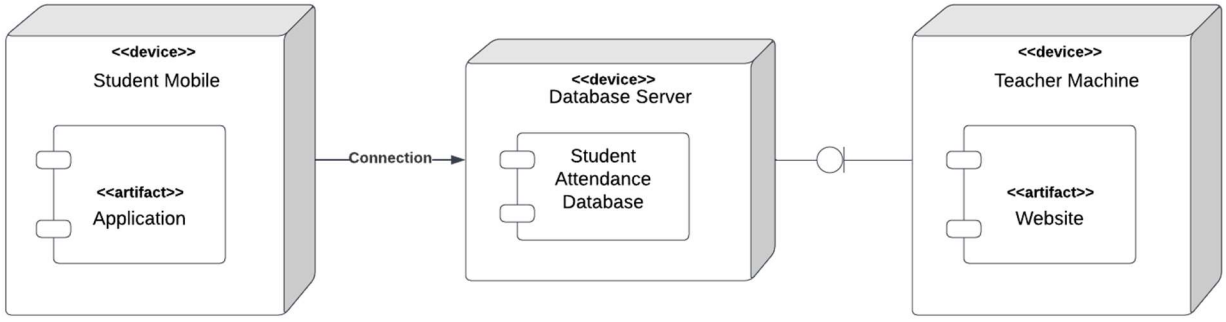
5.3.2 Algorithmic Approaches Used

Below are the steps mentioned for the algorithm used in our project:

1. The first step is to determine the minimum hop required by the flooding algorithm as the beacon node would transmit the data packet in the complete network. The node only keeps the minimum hop value with it and also transmits the data packet to the neighbouring node by increasing the hop value by 1.
2. Next we need to determine the distance between different nodes in the network and the beacon node for that we have used the chebyshev algorithm to calculate the distance between them.
3. The next step is to determine the coordinates of the nodes that are unknown through the least cost square method.

5.3.3 Project Deployment

The below diagram shown is the deployment diagram of our project and as shown the student uses the application in his/her mobile to mark the attendance which after verification goes to the database. The database has an ordered collection of the attendance details by the student which can be used by further analysis by the teachers. Also the teachers can access the simplified view through the website available and through various search queries get the required results.



5.3.4 System Screenshots

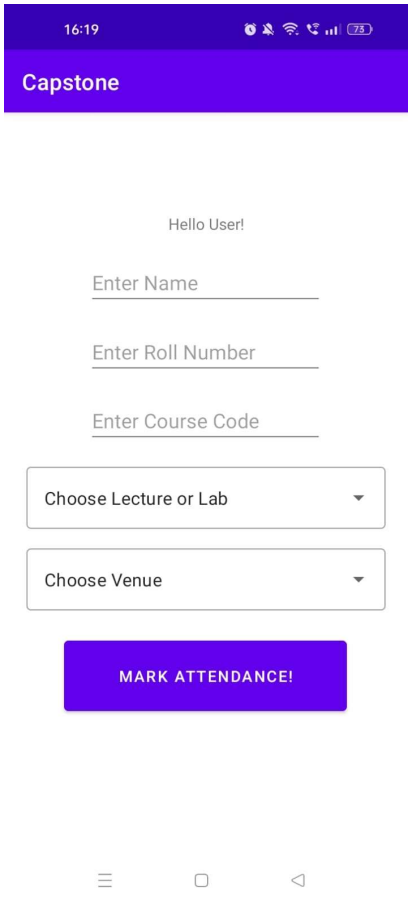


Fig 5.3.4.1 Mobile App Interface

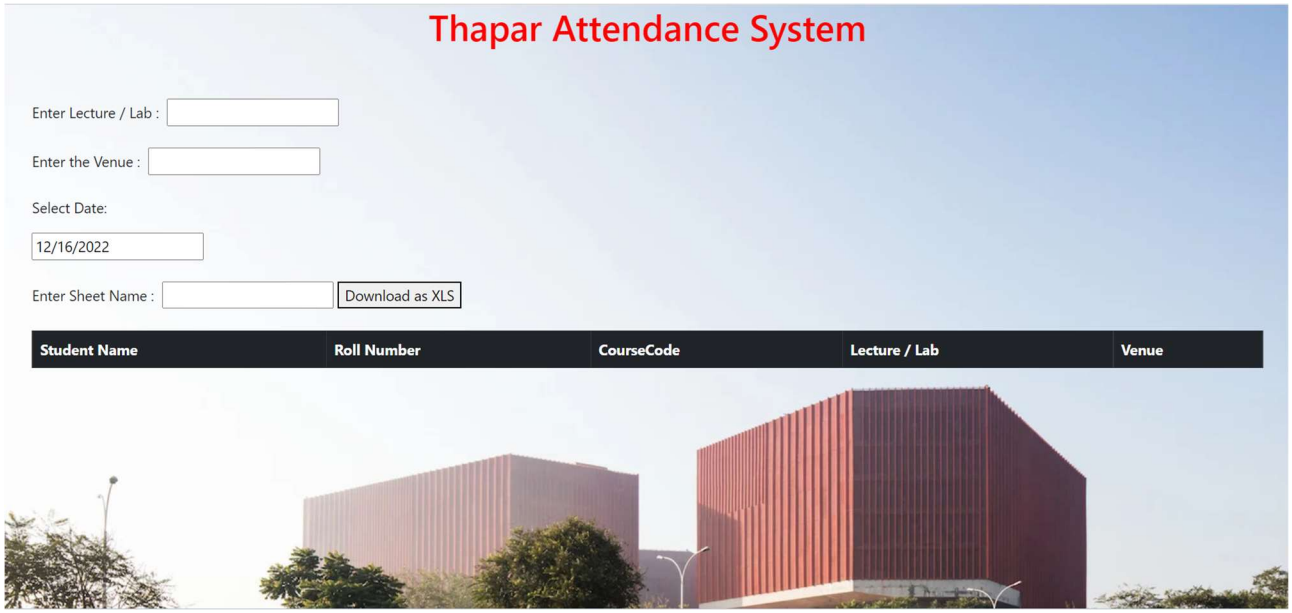


Fig 5.3.4.1 Web Portal Interface

5.4 Testing Process

5.4.1 Test Plan

Testing a SaaS product like ours is very important before releasing it to production. There are various test cases that need to be performed before releasing the product. The purpose of testing is to identify any potential issues with the software, which may have an effect on the production environment.

Different people have different perceptions about what constitutes a successful test case, but one important aspect is that these test cases should be able to uncover major software bugs.

When bugs are not found during testing, it means there are other problems in the business logic or custom code that will be uncovered when released into production. Before every commit, we make to our code and behind every new feature, there is a lot of testing that we do. We test our code on the following parameters:

- Code Flexibility - Is our code flexible? Can it be used by thousands of users without any error.
- Code Reusability - Can our code be reused for any new feature that we create? This helps to reduce development time.
- Code Modularity - Is our code written in modules. It enables to make code faster thus reducing response time.
- Deployment ready - Is our code written for production? Do we need to change anything if our user base grows.

5.4.2 Features to be tested

The features that we would be testing for our application include the following:

S.No.	Feature
1.	User Login
2.	Attendance Marking
3.	Information Retrieval

Table No. 5.4 Features to be Tested

5.4.3 Test Techniques and Strategies

We have employed the following test strategies to test the different functionalities being developed in the project.

1. Unit testing : In this strategy we tested each and every smallest unit of functionality which could be tested.
2. Integration testing : In this phase we integrated the unit functionalities of the previous phase to form a module and then test the module as a whole.
3. System Testing : In this phase we consolidated the constituent modules into a complete package and then tested the package as a whole.

Throughout the different testing phases we had adopted a feedback based approach in which we tried the incorporate the constructive feedback that we gained from the different users who assisted with the testing of the Project.

5.4.4 Test Cases and Results

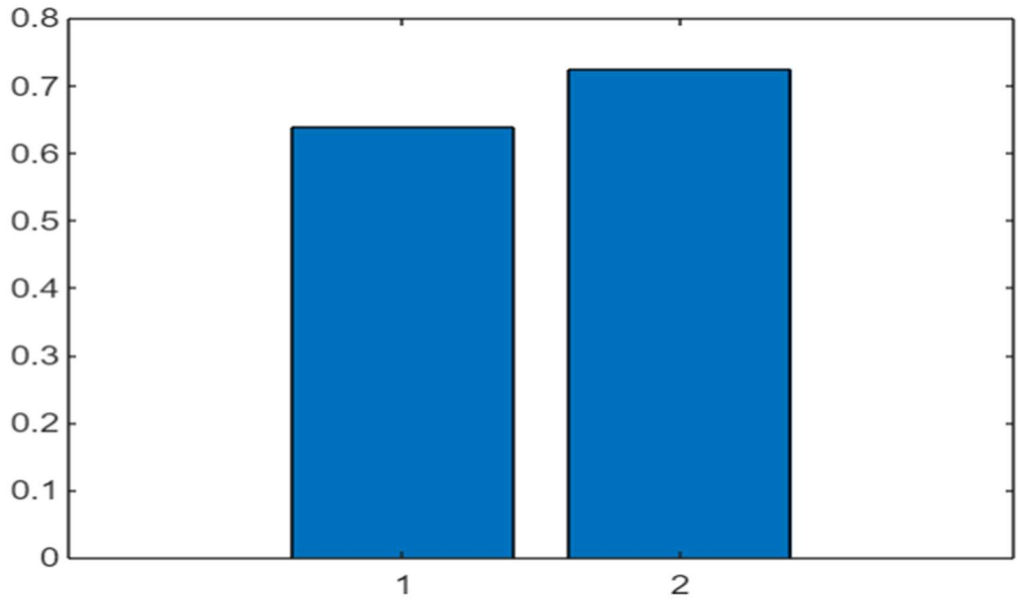
S.No.	Test Case	Input	Expected Output	Actual Output	Result
1.	Login as Teacher	Username / ID , Password	(If password matches) Successful Login	Successful Login	Passed
2.	Login as Student	Username / ID , Password	(If password matches) Successful Login	Successful Login	Passed

3.	Marking the Attendance	Venue , Class Type , Course Code , Roll Number	(If information is correct) Attendance Marked Successfully	Attendance Marked Successfully	Passed
4.	Attendance Display	Class Group , Class Type	Attendance data is Displayed	Attendance data is Displayed	Passed

Table No. 5.5 Test cases and results

5.5 Results and Discussions

We have been able to design a system which showcases an improved DV-Hop algorithm implementation. We were able to achieve this upgrade by analysing the impact and the parameters that the various distance metrics take into account. Through our findings we were able to find that the Chebyshev Distance Metric serves as the optimum metric in the scenario under consideration. We achieved accuracy in 19 out of the 20 random test cases taken into account. We further showcase our results with the help of the below graphical representation.



Where (1) represents the original algorithm and (2) represents the improved algorithm.

5.6 Inferences Drawn

Our project satisfies all the basic functionality as we want but still, there is much to be improved and various new features must be added to make the user experience more pleasant. Moreover, with the proper time, we would like this application to be built for other operating systems.

5.7 Validation of Objectives

Our work completely resonates with the objectives that we laid down from the start. We have been successfully able to provide functionalities which are aligned with our objectives of Security, Accurate Tracking and increasing efficiency. We are providing security using various encryption models. We are achieving accurate tracking of the students through an improved version of the DV-Hop algorithm and with the use of the BSSID concept. We were able to capture improved efficiency in our work by making use of various different distance metrics and finally implementing the most efficient one.

CONCLUSIONS AND FUTURE DIRECTIONS

6.1 Conclusions

There has been rapid development in indoor localization and improvement in various techniques for accurate tracking indoors. One such algorithm is DV Hop, we have made modifications to the original algorithm and improved its accuracy by a significant amount. We have made changes to the various metric calculations and in the distance calculation between adjacent nodes. We had also implemented an attendance tracking system for our Campus, which would help the faculty to have a proper and accurate monitoring of the students. This system uses BSSID of the WiFi Access Points available throughout the campus and the data mapping done of the campus helps to map the student to the correct location.

6.2 Environmental, Economic and Societal Benefits

The Environmental Benefits include:

1. Saving a large amount of paper which was being used to take the attendance.
2. Saving electricity which was earlier being consumed to keep a track of the manually marked attendance by the teachers on their systems.

The Economic Benefits include:

1. Saving a lot of money on the buying of Paper.
2. Faster and accurate tracking saves a lot of time and hassle.

The Social Benefits include:

1. This system can be used in education, industrial, military and medical etc. sectors for the better of the people.
2. It will help to reduce the work of the people and save a lot of time.

6.3 Reflections

We have got a lot of experience and clarity on what sorts of algorithms and techniques are available for location tracking. As we began exploring and learning various strategies we got to know about a lot of available devices and their functioning.

6.4 Future Work

There can be several enhancements that can be done to this attendance monitoring and tracking system. Firstly, there are no coordinates provided which can be done on the basis of various parameters captured by the app. It can also be used for further modifications such as seat location in a room of students and their live tracking and movement. Secondly, there is a scope for further optimisation and more improvement in accuracy that can be done in the portal and algorithm that has been used in the system.

IMPLEMENTATION AND EXPERIMENTAL RESULTS

7.1 Challenges Faced

S. No.	Constraints
1	The mobile application is constrained by the system interface to the system within the mobile phone. Since there are multiple system and multiple manufacturers, the interface will most likely not be the same for every one of them. Also, there may be a difference between what features each of them provide.
2	The Internet connection is also a constraint for the application. Since the application fetches data from the database over the Internet, it is crucial that there is an Internet connection for the application to function.
3	Both the web portal and the mobile application will be constrained by the capacity of the database. Since the database is shared between both application it may be forced to queue incoming requests and therefore increase the time it takes to fetch data.

7.2 Relevant Subjects

S. No.	Subject Code	Subject Name
1.	UCS312	Database Management Systems
2.	UCS412	Software Engineering
3.	UCS414	Computer Networks
4.	UCS301	Data Structure
5.	UCS542	UI & UX Specialist

7.3 Interdisciplinary Knowledge Sharing

Our Project covers different parameters. It consists of knowledge about data security, basic Software Development knowledge, Localisation concept etc. We tried to implement the knowledge in the real world by building an application with the help of the necessary subjects from various disciplines.

7.4 PEER ASSESSMENT MATRIX

		Evaluation of		
		Manan Ghai	Akshat Singh	Kritarth Gupta
Evaluation By	Manan Ghai	4	5	3
	Akshat Singh	3	4	5
	Kritarth Gupta	5	3	4

7.5 Role Playing and Working Schedule

- Akshat Singh- ApplicationDevelopment
- Kritarth Gupta- Web Development
- Manan Ghai- DV Hop Algorithm

7.6 Student Outcomes (SO) Description And Performance Indicators (PI)

S.No	SO Description	Outcome
1.1	Ability to identify and formulate problems related to computational domain	Identified problem of attendance taking and monitoring. The challenge is how to remove the manual system and introduce an automated system.
1.2	Apply engineering, science, and mathematics to obtain analytical, numerical, and statistical solutions to solve engineering problems.	Used mathematical concepts like distance metric and data collection in accurate location tracking in the campus which were also used to interpret and analyze the results.
2.1	Design computing system(s) to address needs in different problem domains & build prototypes, simulations, proof of concepts, wherever necessary, that meet design & implementation specifications.	The final design launched after testing the prototype gave a lot better results and can be used to also implement in the real world for even better understanding of localisation.
2.2	Ability to analyze the economic trade-offs in computing systems.	The economic cost for the project is minimal as it majorly involves the use of cloud. But can be later on extended.
3.1	Prepare and present a variety of documents such as project or laboratory reports according to computing standards and protocols.	Prepared various documents related to our project by referring to different research papers and models.
3.2	Able to communicate effectively with peers in a well organized and logical manner using adequate technical knowledge to solve computational domain problems and issues.	Proper communication in the team was developed to work on the professional aspects of the project.
4.1	Aware of ethical and professional responsibilities while designing and implementing computing solutions and innovations.	Ethics and professionalism were kept in mind to make the project in an appropriate manner.
4.2	Evaluate computational engineering solutions considering environmental, societal, and economic contexts.	Societal aspects were considered in making our project as it is based on an attendance system.
5.1	Participate in the development and selection of ideas to meet established objectives and goals.	Brainstorming was done to select a final idea to work upon.
5.2	Able to plan, share and execute task responsibilities to function effectively by creating a collaborative and inclusive environment in a team.	Regular team meetings were organized to create a team work environment.
6.1	Ability to perform experimentations and further analyze the obtained results.	Model was analyzed to check the accuracy of the obtained results.
6.2	Ability to analyze and interpret data, make necessary judgement(s) and draw conclusions(s).	Research papers were analyzed and looked upon thoroughly.
7.1	Able to explore and utilize resources to enhance self-learning.	Learning through research papers, models and documents were done.

7.7 Brief Analytics Assessment

1. What sources did the group consult to develop a list of issues that could be used as a project?

● Our group has worked on the project as a team. We knew the importance of the capstone project. We had many discussions with our supervisor and other senior faculty along with the literature survey. We did a lot of research on the internet and came with the right ideas to make our project successful.

2. What analytical, computational, and/or experimental methods did your project group use to arrive at a solution to the project's problem?

● Our project has been done analytically on the online tools. We have made our project together in the hostels as well. The values of the components were discussed with our supervisor and the calculated values have been approximated for better results.

3. Did the project require evidence of knowledge of basic, scientific, and/or engineering principles? If yes, when did you apply?

● Yes, the project totally gave us the opportunity to use our basic Computer fundamentals covering many subjects like Database Management Systems, Software Engineering, Computer Networks, Data Structure and UI & UX Specialist.

4. How did your group share responsibilities, share scheduling information with other members of the team, and coordinate design and manufacturing dependencies?

● We have a group of 3 members and we share a great amount of coordination and unity. Apart from being a group, we all are good friends too. We communicate regarding our project in the institute's library after class and on WhatsApp groups as well. We've been meeting our mentor on a regular basis that has helped us solve our queries and work under proper guidance.

5. What resources did you use during the course of the project to learn new material that was not taught in class?

● We made use of various online resources and websites like YouTube etc. to study about the concepts of localisation and also about enhancing our skills for both Web development as well as Mobile App development.

6. Does this project inspire you to use technology to solve real-world problems?

● Being a Computer Engineer doesn't mean just having theoretical knowledge but it's about how we implement it in real life. We have made a Web Application that has helped us analyze real world problems and solve them with a better approach with even more efficiency.
Our project has a lot of capabilities to work in real life scenarios and can be of a great use to people.

7. List the software tools used to analyze various problems in the capstone project.

● Android Studio
● VS Code
● Selenium
● Lucid Chart

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